Homework 24: 9.7-9.9 **Name** _____
AP Calculus BC **Due** _____
Period _____ **Score** _____ **/ 100**

Skill Builder 9.7: Defining Polar Coordinates & Differentiating in Polar Form

1. For each of the following polar coordinates, find the corresponding rectangular coordinates.

a.) $\left(6, \frac{\pi}{2}\right)$	**b.)** $\left(1, \frac{7\pi}{4}\right)$	**c.)** $\left(-1, \frac{-7\pi}{4}\right)$	**d.)** $\left(4, \frac{-\pi}{3}\right)$

2. For each of the following rectangular coordinates, find the corresponding polar coordinates, one with a positive radius and the other with a negative radius.

a.) $(-3,3)$	**b.)** $\left(\frac{1}{2}, \frac{-\sqrt{3}}{2}\right)$	**c.)** $(0,-4)$	**d.)** $\left(-3\sqrt{3},-3\right)$

3. For each of the following rectangular equations, change it to polar form and confirm using your calculator.

a.) $5x - y = 7$	**b.)** $xy = 16$
c.) $(x-1)^2 + y^2 = 1$	**d.)** $\left(x^2 + y^2\right)^2 = \left(x^2 - y^2\right)^2 + 1$

4. For each of the following polar equations, change it to rectangular form and confirm using your calculator. Solve for y for each answer.

a.) $r = 4$	**b.)** $\tan^2\theta = 9$
c.) $r = 4\sin\theta$	**d.)** $r = \dfrac{1}{1-\cos\theta}$

5. For $r = 2 + 3\sin\theta$, find $\dfrac{dy}{dx}$ and the slope of the tangent line at the following polar points.

Work for $\dfrac{dy}{dx}$:

a.) $\left(5, \dfrac{5\pi}{2}\right)$	**b.)** $(2, \pi)$
c.) $\left(-1, \dfrac{3\pi}{2}\right)$	**d.)** $\left(\dfrac{7}{2}, \dfrac{\pi}{6}\right)$

6. For each of the following, find the points of horizontal and vertical tangency (if any).

a.) $r = 4(1+\cos\theta)$

b.) $r = \sin\theta\cos^2\theta, \quad 0 \le \theta \le \pi$

Skill Builder 9.8: Finding the Area of a Region Bounded by a Single Polar Curve

In the following, graph on your calculator and find the area of the region. Set up and use your TI-Nspire.

1.) One petal of $r = 6\cos 3\theta$	**2.)** One petal of $r = 2\cos 2\theta$
3.) One petal of $r = \sin 5\theta$	**4.)** Interior of $r = 2 - \sin\theta$ (above the polar axis)
5.) Inner loop of $r = 1 - 2\cos\theta$	**6.)** Between the loops of $r = 1 + 2\cos\theta$

Find the points of intersection of the graphs of the following. Show work and only use the calculator to verify.

7.) $r = 2 + \cos\theta$ and $r = 2 - \cos\theta$	**8.)** $r = 1 + \cos\theta$ and $r = 1 - \sin\theta$
9.) $r = 1 + \cos\theta$ and $r = 3\cos\theta$	**10.)** $r = 4\sin 2\theta$ and $r = 2$

Skill Builder 9.9: Finding the Area of a Region Bounded by Two Polar Curves

Use your calculators to both graph the polar equations and find the area of the indicated region.

1.) Common interior of $r = 4(1+\sin\theta)$ and $r = 4(1-\sin\theta)$	**2.)** Outside $r = 1+\sin\theta$ and inside $r = 2$
3.) Inside the lemniscate of $r^2 = 8\cos 2\theta$ and outside $r = 2$	**4.)** The sonar signal from a submarine is modeled by $r = a\cos^2\theta$. Find the area of the region between the two curves for $a = 5$ and $a = 10$

MCQ Enrichment – Unit 9B Polar Equations and Calculus

For each of the following problems, do not use your calculator unless you see the calculator icon.
Credit will only be given for correct answers with adequate supporting work.

1. Convert the rectangular equation $y^2 = 9x$ to polar form.

(A) $r = 3\csc^2\theta\cos\theta$ **(B)** $r = \dfrac{3\cos\theta}{\sin\theta}$

(C) $r = 9\csc^2\theta\cos\theta$ **(D)** $r = 9\sin^2\theta\cos\theta$

2. Convert the polar equation $r = \sec\theta + \csc\theta$ to rectangular form.

(A) $y = -\dfrac{1}{x}$ **(B)** $y = \dfrac{1}{x}$

(C) $y = \dfrac{x}{x-1}$ **(D)** $y = \dfrac{x}{x+1}$

3. For the polar curve $r = 3 - 4\sin\theta$, find $\dfrac{dy}{dx}$ at $\theta = 0$.

(A) $\dfrac{-3}{4}$ **(B)** $\dfrac{3}{4}$ **(C)** $\dfrac{-4}{3}$ **(D)** $\dfrac{4}{3}$

4. Find the equation of the line tangent to $r = 2 + 3\sin\theta$ at $\theta = \pi$.

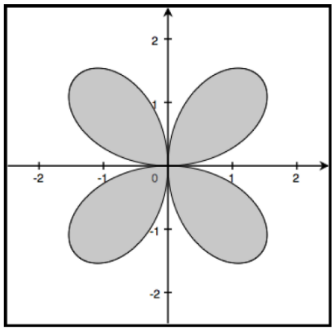
(A) $y = \dfrac{2}{3}(x-2)$ **(B)** $y = -\dfrac{2}{3}(x+2)$

(C) $y = -\dfrac{3}{2}(x+2)$ **(D)** $y = \dfrac{3}{2}(x-2)$

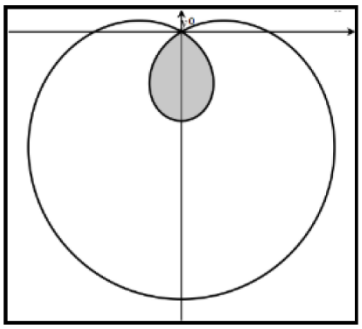
5. The graph of the four-leaf rose $r = 2\sin 2\theta$ is shown in the figure to the right. Which of the following gives the correct area of the figure?

I. $2\displaystyle\int_0^{2\pi} \sin^2 2\theta\, d\theta$ II. $4\displaystyle\int_0^{\pi} \sin^2 2\theta\, d\theta$ III. $8\displaystyle\int_0^{\pi/2} \sin^2 2\theta\, d\theta$

(A) I and II only **(B)** I and III only
(C) II and III only **(D)** I, II and III



6. The graph of the limaçon $r = 2 - 4\sin\theta$ is shown in the figure to the right. Which of the following gives the area of the shaded inner loop?



I. $\int_0^{\pi/6} (2-4\sin\theta)^2\, d\theta$ **II.** $\int_{\pi/6}^{\pi/2} (2-4\sin\theta)^2\, d\theta$

III. $\int_{3\pi/2}^{2\pi} (2-4\sin\theta)^2\, d\theta$ **IV.** $\int_0^{\pi/2} (2-4\sin\theta)^2\, d\theta - \int_0^{\pi/6} (2-4\sin\theta)^2\, d\theta$

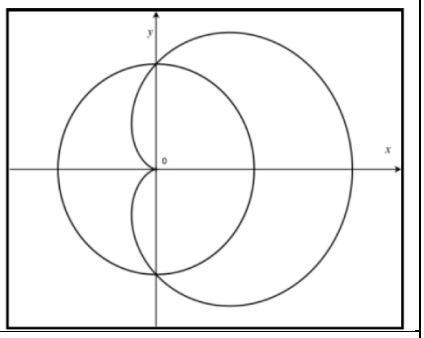
(A) I and II only **(B)** II only **(C)** II and III only **(D)** II and IV only **(E)** I, II, and III only

(F) III only **(G)** I, II, III and IV **(H)** none correctly express the area of the inner loop

7. Find the area common to both the circle of radius 1 and the cardioid $r = 1 + \cos\theta$ as shown in the figure to the right. Write the integral expression.

(A) 0.356 **(B)** 1.215

(C) 1.571 **(D)** 1.927



8. A particle moving along the polar curve $r = e^{\sin\theta}$ has position $(x(t), y(t))$ at time *t* with $\theta = 0$ corresponding to $t = 0$. The particle moves along the curve such that $\frac{dr}{dt} = \frac{d\theta}{dt}$. Describe the motion of the particle at $t = \frac{5\pi}{4}$.

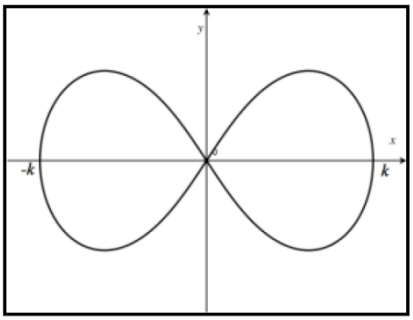
I. Getting closer to the *x*-axis **II.** Getting closer to the *y*-axis **III.** Getting closer to the origin

(A) I only **(B)** II only **(C)** I and II only **(D)** II and III only

9. For how many values of θ does the curve $r = 4(1+\sin\theta)$ have horizontal tangents on $0 \le \theta \le 2\pi$?

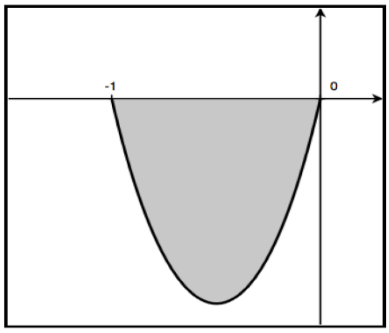
(A) One **(B)** Two **(C)** Three **(D)** Four

10. The graph of the lemniscate $r = k\sqrt{\cos 2\theta}$ where *k* is a constant is shown in the figure to the right. Find the area bounded by the curve.



(A) $\frac{k^2}{4}$ **(B)** $\frac{k^2}{2}$ **(C)** k^2 **(D)** $2k^2$

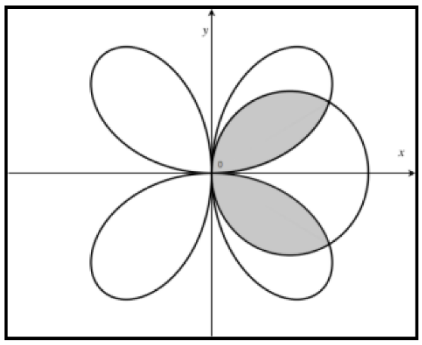
11. Find the area enclosed by $r = \tan\theta - 1$ between $\theta = 0$ and $\theta = \frac{\pi}{4}$ as shown in the figure to the right.



(A) $\frac{1}{2} + \ln 2$ **(B)** $\frac{1}{2} + 2\ln 2$

(C) $\frac{1 - \ln 2}{2}$ **(D)** $1 - \frac{\ln 2}{2}$

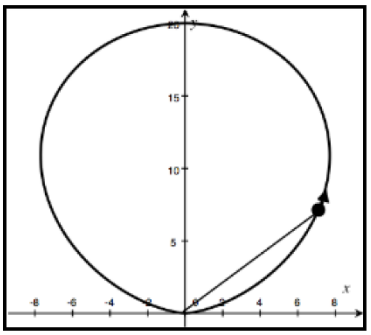
12. To find the shaded area enclosed by the rose curve $r = 2\sin\theta\cos\theta$ a and the circle $r = \cos\theta$ as shown by the figure to the right, we find the first quadrant area and double it. The resulting calculation is the sum of two integrals, one from row M and the other from row N. What are they?



	1	2	3
M	$\int_0^{\pi 6} \left(4\sin^2\theta\cos^2\theta\right)d\theta$	$\int_{\pi/6}^{\pi} \left(4\sin^2\theta\cos^2\theta\right)d\theta$	$\int_{\pi/6}^{\pi/2} \left(4\sin^2\theta\cos^2\theta\right)d\theta$
N	$\int_{\pi/6}^{\pi} \cos^2\theta d\theta$	$\int_{\pi/6}^{\pi/2} \cos^2\theta d\theta$	$\int_0^{\pi/6} \cos^2\theta d\theta$

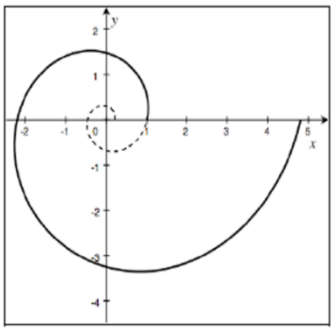
(A) M1, N1 **(B)** M1, N2 **(C)** M2, N1 **(D)** M3, N3

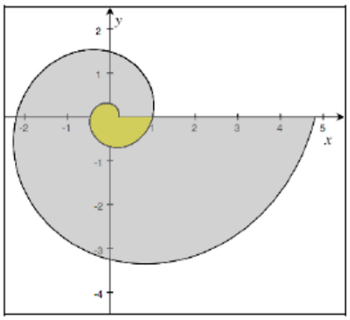
13. A child's amusement park ride has a car that follows the track of $r = 20\sin^2\theta$ as shown in the figure to the right, where *x* and *y* are measured in feet. The angle changes at $\frac{10^\circ}{\text{sec}}$. How fast is the car moving along the track in ft/sec at the time it is at the point on the track where it makes a 45° angle to the horizontal?



(A) $\frac{50\pi}{3}$ **(B)** $\frac{5\pi\sqrt{5}}{9}$ **(C)** $100\sqrt{5}$ **(D)** $\frac{10\pi}{9}$

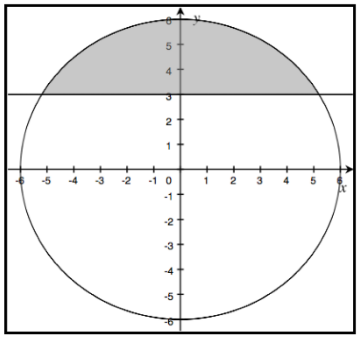
14. The graph of the logarithmic spiral $r = e^{\theta/4}$ is shown in the figure below to the left. The spiral from $\theta = 0$ to $\theta = 2\pi$ is solid while from $\theta = -2\pi$ to $\theta = 0$ is dashed. Find the area of the grey shaded region in the figure below to the right.





(A) $e^{\pi} + e^{-\pi} - 2$ **(B)** $e^{\pi} + e^{-\pi}$ **(C)** $e^{\pi} - e^{-\pi}$ **(D)** $e^{\pi} - e^{-\pi} - 2$

15. A 12-inch diameter pizza is shown in the figure to the right. Bernie D. Rolz and his wife share the pizza with the wife getting the shaded region. Polar equations are used to find the area between the polar curves $r = 6$ and $r = 3\csc\theta$ (which graphs a horizontal line). Find the area of the shaded region.



(A) $24\pi - 18\sqrt{3}$ **(B)** $12\pi - 3\sqrt{3}$

(C) $12\pi - 9\sqrt{3}$ **(D)** $36\pi - 54\sqrt{3}$

16. A particle travels along the polar curve $r = 2\theta - 4\cos\theta$ for $0 \le \theta \le 2\pi$. What is the greatest distance of the particle from the origin? Show your work.

(A) 5.760 **(B)** 8.055
(C) 8.566 **(D)** 10.794